## Question 1

Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**For example:**

| Input | Result |
|---|---|
| 9 | 12 |

**Answer:** (penalty regime: 0 %)

```
#include <stdio.h>

void function(int n)
{
    int count = 0;

    int i = 1;
    count++;

    int s = 1;
    count++;

    while (s <= n)
    {
        count++;

        i++;
        count++;

        s += i;
        count++;
    }

    count++;

    printf("%d", count);
}

int main()
{
    int n;
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 2 | 12 | 12 | ✔ |
| ✔ | 1000 | 5002 | 5002 | ✔ |
| ✔ | 143 | 717 | 717 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

## Question 1

Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```c
#include <stdio.h>
int main()
{
    int n,count=0;
    scanf("%d",&n);
    count=2*n+1;
    for(int i=1;i<=n;i++)
    {
        if(n%i==0)
        {
            count++;
        }
    }
    printf("%d",count);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 31 | 31 | ✔ |
| ✔ | 25 | 54 | 54 | ✔ |
| ✔ | 4 | 12 | 12 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

## Question 1

Incorrect Mark 0.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include <stdio.h>

void function(int n) {
    int c = 0;
    for (int i = n / 2; i < n; i++) {
        for (int j = 1; j < n; j = 2 * j) {
            for (int k = 1; k < n; k = k * 2) {
                c++;
            }
        }
    }
    printf("Counter c = %d\n", c);
}

int main() {
    function(16);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✗ | 4 | 30 | Counter c = 128 | ✗ |
| ✗ | 10 | 212 | Counter c = 128 | ✗ |

## Question 1

Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```c
#include <stdio.h>
int main()
{
    int n,rev=0,remainder,count=0;
    scanf("%d",&n);

    while(n!=0)
    {
        remainder=n%10;
        rev=rev*10+remainder;
        n/=10;
        count+=4;
    }

    count+=3;
    printf("%d",count);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 11 | 11 | ✔ |
| ✔ | 1234 | 19 | 19 | ✔ |